

THE HISTORY OF CONSTELLATIONS

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THE FIRST CONSTELLATIONS were patterns of stars that ancient peoples employed for navigation, timekeeping, and storytelling. Recently, the pictorial aspect of constellations has become less significant, and they have become simply delineated regions of the sky, although the attraction of the myths and legends remains.

EARLY CONSTELLATION LORE

The constellation system used today stems from patterns recognized by ancient Greek and Roman civilization. The earliest surviving account of ancient Greek constellations comes from the poet Aratus of Soli (c. 315–c. 245 BCE). His poem, the *Phaenomena*, written around 275 BCE, describes the sky in storybook fashion and identifies 47 constellations. It is based on a lost book of the same name by the Greek astronomer Eudoxus (c. 390–c. 340 BCE). Eudoxus reputedly introduced the constellations to the Greeks after learning them from priests in Egypt. These constellations had been adopted from Babylonian culture; they were originally created by the Sumerians around 2,000 BCE. However, the Greeks attached their own myths to the constellations detailed by Eudoxus, and Aratus's storybook of the stars proved immensely popular. Sometime in the 2nd century CE, it was joined by a more elaborate work of constellation lore called *Poetic Astronomy*, written by the Roman author Hyginus. Many editions of both these works were produced and translated over the centuries.

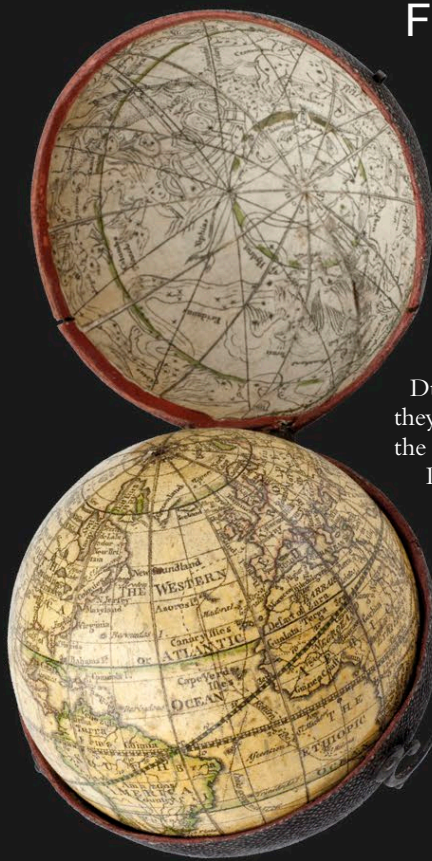
ANTICANIS

This page from a 9th-century edition of the star myths of Hyginus shows the constellation Canis Minor, here termed Anticanis. Hyginus's words, in Latin, form the shape of the dog's body.



FILLING THE HEAVENLY SPHERE

The oldest surviving star catalog dates from the 2nd century CE and is contained in a book called the *Almagest*, written by the Greek astronomer and geographer Ptolemy (see panel, opposite). It records the positions and brightnesses of one thousand stars, arranged into 48 constellations, based on an earlier catalog by Hipparchus of Nicaea (c. 190–c. 120 BCE). In the 10th century CE, an Arab astronomer, al-Sufi (see p.421), updated the *Almagest* in his *Book of Fixed Stars*, which included Arabic names for many stars. These Arabic names are still used today, although often in corrupted form. No more constellations were introduced until the end of the 16th century, when Dutch explorers sailed to the East Indies. From there, they could observe the southern sky that was below the European horizon. Two navigators, Pieter Dirkszoon Keyser and Frederick de Houtman (see p.416), cataloged nearly 200 new southern stars, from which they and their mentor, Petrus Plancius (see p.358), a leading Dutch cartographer, created 12 new constellations. Plancius also created other northern constellations, forming them between those listed by Ptolemy. Nearly a century later, Johannes Hevelius (see p.384), a Polish astronomer, filled the remaining gaps in the northern sky, and in the mid-18th century, the French astronomer Nicolas Louis de Lacaille (see p.422) introduced another 14 constellations in the southern sky.



POCKET GLOBE

This pocket globe from the Science Museum, England, positions the Earth within a shell that depicts the surrounding celestial sphere. On the inside of the open shell are the constellations, drawn as mirror images.



GLOBAL COVERAGE

This beautiful celestial globe was made in 1692 by Vincenzo Coronelli, a celebrated Italian globe maker. As in all celestial globes, the constellation figures are shown reversed by comparison with their appearance in the sky.

